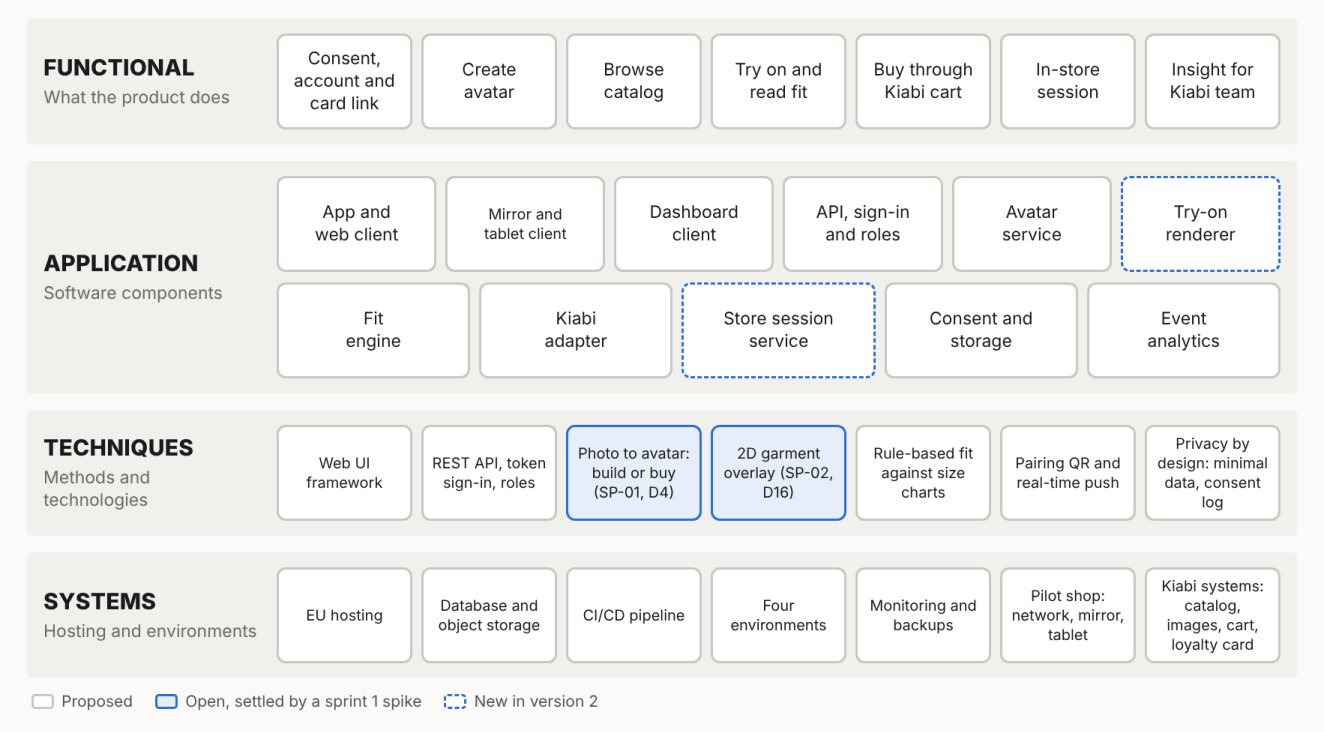
Checkpoint 3a. Target architecture

Project: Fittybird x Kiabi, 360° Framework
Checkpoint: 3a, Period 3 (technical building blocks)
Version: 2. Changes from version 1 are listed in section 13.
Terms: `shared/glossary.docx`

The architecture is organized in the four categories the framework asks for: **Systems**, **Techniques**, **Application** and **Functional**. Each category answers a different question, and every element is classified in exactly one of them.

Category	The question it answers	What goes in it
Functional	What does the product do for its users?	Business capabilities and the user-facing functions
Application	Which software pieces deliver those functions?	Components, interfaces, clients
Techniques	Which methods and technologies build those pieces?	Languages, frameworks, models, protocols, design rules
Systems	Where does it run, and how is it operated?	Hosting, networks, environments, pipelines, operations, Kiabi's own systems



Status words used below follow the decision log (`C-backlog/decision-log.docx`): **Confirmed** (the team has stated it), **Proposed** (the tech lead's working choice, to confirm at the checkpoint review) and **Open** (a decision is still to be made, with the sprint or spike that settles it). Decision numbers D4 to D20 refer to that log.

1. Functional architecture

Function	What it does	Backlog	Persona
Consent and account	Records consent, signs the shopper in with an account, links the loyalty card to the account once, deletes data on request	US-01, US-02, US-03, US-23	Nadia, Karim
Create avatar	Guides photo capture, builds the avatar, confirms height and usual size, checks photo quality	US-04, US-05, US-06, US-11	Léa
Browse catalog	Shows Kiabi articles with filters	US-07	Lucas
Try on and read fit	Draws an article on the avatar and rates each body zone	US-08, US-09, US-10, US-16	Léa
Buy through Kiabi cart	Hands the chosen article and size to Kiabi's cart and checkout	US-12	Lucas
In-store session	Pairs the shopper's phone with the mirror, loads the avatar, sends size requests to a staff tablet	US-13, US-14, US-17	Karim, Inès
Feedback and measurement	Captures fit feedback and return reasons, records events	US-15, US-22	Claire
Insight for Kiabi	Shows usage, size choices and cross-channel sessions	US-18	Claire

2. Application architecture

Component	Role	Talks to	Status
App and web client	Shopper interface on phone and browser	API	Proposed
Mirror and tablet client	In-store screen and staff tablet. The same web application in a different layout	API	Proposed
Dashboard client	Interface for the Kiabi team	API	Proposed
API and sign-in	Single entry point for every client. Handles authentication, roles (shopper, staff, Kiabi team) and routing	All services	Proposed
Avatar service	Turns photos into an avatar. Photos are deleted once the avatar is made	API, storage	Open (SP-01, sprint 1)
Try-on renderer (new)	Draws a garment on the avatar from catalog images, front and back views	Avatar service, Kiabi adapter	Open (SP-02, sprint 1)
Fit engine	Compares the avatar with a size chart and returns the rating per zone	Avatar service, Kiabi adapter	Proposed (D8)
Kiabi adapter	The only door to Kiabi systems: catalog, size charts, garment images, cart hand-off, loyalty-card check. Runs against a mock feed until Kiabi gives access	Kiabi systems	Proposed
Store session service (new)	Pairs a mirror with a shopper's phone, holds the session, relays size requests to the staff tablet in real time	API, mirror and tablet clients	Proposed (D17, D20)
Consent and storage	Stores avatars, consent log and profile data	API	Proposed
Event analytics	Receives usage and feedback events and serves aggregates to the dashboard	API	Proposed

3. Techniques

Area	Choice	Why	Status
Client	Web application framework, responsive from phone to tablet to mirror. Touch targets of 44 px or more	One codebase for three surfaces. Framework to be chosen by the UI lead in sprint 1 (React is the working assumption)	Proposed
API	REST with JSON, token-based sign-in, roles for shopper, staff and Kiabi team	Simple, testable, easy to hand to a development team	Proposed
Photo to avatar	Build or buy. Compare three options on the same 10 sample photo sets: quality, cost, speed, privacy and where the provider processes photos	The largest technical unknown (risks R6, R7, R16)	Open, decided at the end of sprint 1 (D4)
Try-on rendering (new)	Version 1: a 2D garment overlay drawn from Kiabi catalog images, for adult tops first. No 3D garment models	The fit zones in the glossary (shoulders, chest, waist, sleeve length) are tops zones. A simple overlay answers risk R8 and fits the 8 days of US-08	Open, decided by SP-02 (D16)
Fit calculation	Rule-based comparison between avatar measurements and the size chart, shown as a range with a size confirmation step	Explainable, needs no training data, protects trust (risk R3)	Proposed (D7, D8)
Data minimization	Photos deleted once the avatar is made, unless the shopper opts in to keep them	Answers risks R1 and R9	Proposed (D5)
Consent	Versioned consent log with date and withdrawal date	Proves what the shopper agreed to	Proposed
Identity in store (changed)	The mirror shows a pairing QR code. The shopper's signed-in phone scans it and authorizes the mirror to load their avatar. The QR code stands in for the NFC tap in the first pilot	No camera or card reader on the mirror. Removes the integration risk R10 from the critical path	Proposed (D6, D17)
Loyalty card link (new)	Linked once to the account by card number, checked through the Kiabi adapter. If Kiabi cannot check a card number, the link waits and the pairing QR is the only way in	Carries the cross-channel promise (growth lever L1). Depends on the Kiabi answer to OPS-01	Proposed (D18). Kiabi answer open
Real-time requests (new)	Server-pushed events from the store session service to the staff tablet, with a polling fallback. The staff tablet signs in with a shop-level role in the pilot	A size request has to reach the associate in seconds (US-14)	Proposed (D20)
Quality	Automated tests on the API and the fit engine, code review by a second person	Backs the Definition of Done	Proposed
Measurement	Event schema defined before the first pilot, with a control group. No photos in events	Needed to prove the return-rate effect	Proposed (US-22, D14)

4. Systems

Area	Choice	Why	Status
Hosting	A cloud region in the European Union. Provider chosen in sprint 1	Personal data of French shoppers	Proposed (D8). Provider open
Data stores	A relational database for profiles, consent and sessions. Object storage for avatars	Different data, different retention	Proposed
Environments	Four: Local, Dev, Staging, Production (see section 7)	Safe path from laptop to pilot	Proposed
Delivery pipeline	Automated build, test and deploy on every merge	Keeps staging current for reviews	Proposed. Tool depends on the repository host
Security	Encrypted connections, encryption at rest, secrets kept out of the repository, role-based access, no real photos outside Production	Body photos are sensitive	Proposed
Operations	Monitoring, alerts, daily backups. Backups are kept for a fixed period, so deleted data leaves them too	A pilot with real shoppers, and the right to delete	Proposed. Period set in the privacy review
Devices	One in-store screen for the mirror, one staff tablet	Pilot shop only	Open (OPS-02, sprint 4)
Shop network (new)	Check Wi-Fi coverage at the pilot shop before sprint 4	The mirror and tablet need a steady connection (risk R17)	Open (OPS-02)
Kiabi systems	Catalog and size charts, garment images, e-commerce cart, loyalty card with NFC tag	Owned by Kiabi. Access via the Kiabi adapter	To obtain from Kiabi (OPS-01)

5. Players

Player	Role
Shopper	Creates an avatar, tries items on, buys
Store associate	Receives size requests on the tablet and brings the item
Kiabi e-commerce and marketing team	Reads the dashboard, owns catalog and cart
Kiabi IT and data	Gives access to catalog, size charts, garment images, cart and loyalty systems
Kiabi privacy contact (new, to be named)	Confirms the legal basis, retention periods and the privacy review before the pilot
Pilot shop manager (new)	Agrees to the pilot, hosts the mirror and the tablet (OPS-02)
Product team	Elie Zaylaa, Boyu Fu, Muhammad Rao, Hadeed Rauf: build and run the platform
Avatar technology provider	Only if sprint 1 decides to buy (SP-01). It would process photos on our behalf, so it needs a data processing agreement and processing in the EU
School jury	Reviews each checkpoint on Gandalf

6. Data flows

Step	Flow	Category touched
1	The shopper consents in the app, then sends photos. The API passes them to the avatar service, which returns the avatar. Photos are deleted afterwards (D5)	Functional, Application, Techniques
2	The avatar, the confirmed height and size and the consent log are stored	Application, Systems
3	The Kiabi adapter pulls catalog items, size charts and garment images. The fit engine compares the avatar with each size chart	Application, Techniques
4	The try-on renderer draws the garment on the avatar and the app shows the fit per zone. The shopper picks a size and the adapter hands the article to Kiabi's cart	Functional, Application

Step	Flow	Category touched
5	The shopper enters the loyalty-card number once. The adapter checks it with Kiabi and the API links it to the account (D18)	Functional, Application
6	In store, the mirror shows a pairing QR code. The signed-in phone scans it, the store session service confirms the pairing to the API and the mirror loads the same avatar (D17)	Functional, Systems
7	A size request goes from the mirror to the store session service, which pushes it to the staff tablet with the article and size (and the fit view, US-17)	Application
8	When the shopper taps delete, the API removes the avatar, any kept photos, the measurements and the card link within 24 hours, and records the withdrawal date. Backups drop it when their retention period ends	Functional, Application, Systems
9	Every step sends an event to analytics, without photos or measurements. The dashboard reads aggregates only	Application, Systems

7. Environments

Environment	Purpose	Data
Local	Development on each laptop	Fake photos, sample catalog
Dev	Shared integration, deployed on every merge	Synthetic data
Staging	Sprint reviews and demos, same set-up as production	Test accounts, sample Kiabi feed. Photos of consenting team members only
Production	Pilot with real shoppers on the web and in one shop	Real data, EU hosting, backups, consent log

Rule: photos of real shoppers exist only in Production, and only after the privacy review (D10).

8. Data stored and retention

The periods marked "to confirm" are proposals. The Kiabi privacy contact sets the final values in the privacy review (PRIV-01, PRIV-02).

Data	Where	Kept	Deleted when
Photos	Temporary object storage	Only while the avatar is built, unless the shopper opts in to keep them	Right after the avatar is made, or on request (D5)
Avatar and measurements	Object storage and database	While the account is active	The shopper deletes them (within 24 hours, US-02), or after 12 months without use (to confirm)
Account and loyalty-card link	Database	While the account is active	With the account
Consent log	Database	Date, version and withdrawal date. No photos	Period set by the Kiabi privacy contact (open)
Pairing and size-request sessions	Database or cache	A few minutes to a shop visit	On expiry: pairing code valid for 2 minutes, session closed after 30 minutes without use (to confirm)
Events	Analytics store	Pseudonymous identifier. No photos, no measurements	Period set in the privacy review (open)
Backups	Backup storage	Daily, for a fixed period	On schedule, so a deleted avatar leaves them too

9. Quality targets

Requirement	Target	Source
Avatar ready after valid photos	Within 60 seconds (a proposal to test in sprint 2)	US-05
Time to first avatar	Median under 3 minutes	Manifesto section 7

Requirement	Target	Source
Usable avatars	80% of test photo sets by the end of sprint 2	Methodology indicators, R7
Size match	80% of testers get an advised size equal to their real size, from sprint 3	Methodology indicators, R3
Deletion on request	Within 24 hours	US-02
Size request reaches the tablet	Within 5 seconds in the pilot shop (a proposal to test in sprint 4)	US-14
Accessibility	Touch targets of 44 px or more	Growth levers, macro trend 1
Pilot operation	Monitoring with alerts during the pilot weeks. No formal availability promise	Section 4

10. Technical backlog items

Version 1 of this file described components that no backlog item built, and the roadmap's 91.5 person-days cover user stories only. These items close that gap. Sizes use the roadmap scale (S about 1.5 person-days, M about 4, L about 8). They are proposals for the product manager, who owns the backlog (D15).

ID	Item	Answers	Size	Sprint
TECH-01	Hosting, four environments and secrets	Section 4, section 7	M, 4	1
TECH-03	Kiabi adapter core, with a mock feed	R2, section 2	M, 4	1
SP-02	Spike: try-on rendering approach on 5 pilot articles	R8, D16	S, 1.5	1
PRIV-01	Privacy checklist and consent text, before anyone outside the team uploads a photo	R9, D5, D19	S, 1.5	1
TECH-02	Delivery pipeline: build, test, deploy on every merge	Section 4	M, 4	2
US-23	As Karim, I want to link my loyalty card to my account once, so that my avatar follows me into the shop	D18, lever L1	M, 4	4
TECH-04	Monitoring, alerts and backups	Section 4	S, 1.5	5
PRIV-02	Full privacy review before the pilot (decision D10, not estimated in version 1)	R9, D10	M, 4	5

Capacity. The eight items add 24.5 person-days. With the roadmap's 91.5, the plan becomes 116 of the 120 available in sprints 1 to 5 (97%). Loads by sprint become 23.5, 21.5, 24, 24 and 23 against 24, and the slack falls from 28.5 to 4 person-days. Proposed answer: move US-16 (compare two sizes, a Should of 4 person-days) to Later. Slack returns to 8 person-days (7%) and sprint 5 falls to 19. The decision belongs to the product manager. The staff sign-in and push delivery for the tablet (D20) are added to the acceptance criteria of US-14 and may resize it.

11. Technical risks

The risks below are the technical ones from `3b-risks/risk-management.docx`, plus four new ones proposed for the first arbitration session. Scores come from that session, not from this file. Opening scores are likelihood x impact on the 1 to 3 scale in 3b.

ID	Risk	Trigger	Opening	Response and mitigation	Owner
R3	Wrong fit advice raises returns instead of cutting them	Size match under 80% in sprint 3 tests	2x3 = 6	Reduce. Fit as a range, size confirmation step, advice measured against real sizes (D7)	Muhammad
R6	No team member has deep 3D or computer-vision experience	SP-01 shows a gap	3x3 = 9	Reduce. Spike on existing models and services before building (D4)	Muhammad

ID	Risk	Trigger	Opening	Response and mitigation	Owner
R7	Avatar looks inaccurate or fails on poor photos	Usable avatars under 80% of test sets at the sprint 2 demo	3x3 = 9	Reduce. Photo quality check, tips during capture, manual size correction	Muhammad
R8	Garment rendering looks unrealistic on the avatar	Test users call the try-on "fake"	2x2 = 4	Reduce. 2D overlay and a short list of garment categories (SP-02, D16)	Boyu
R9	Photos and body data fall under privacy rules and are sensitive	A legal question the team cannot answer	2x3 = 6	Avoid. Privacy by design, EU hosting, consent log, retention limits, review before the pilot (D5, D10, D19)	Muhammad
R10	NFC reading in store is hard to integrate	Kiabi cannot expose card identity	2x2 = 4	Avoid. Pairing QR code in the pilot (D6, D17)	Muhammad
R15 (new)	Kiabi cannot supply garment images usable for try-on	No usable images for 20 pilot articles by the end of sprint 1, week 2	2x3 = 6	Reduce. Ask in OPS-01. Fallback: 20 hand-prepared articles	Muhammad
R16 (new)	A bought avatar provider processes photos outside the EU or without a data processing agreement	A shortlisted provider cannot state where it processes photos	2x3 = 6	Avoid. Only providers with EU processing and an agreement. Otherwise build	Muhammad
R17 (new)	The pilot shop network is too weak for the mirror and the tablet	The signal test in the pilot shop fails	2x2 = 4	Reduce. Test before sprint 4, polling fallback. Last fallback: staff-only demo (R14)	Muhammad, with Hadeed
R18 (new)	Technical enablers use the capacity planned for stories	Velocity under 20 person-days in sprint 1 or 2	3x2 = 6	Reduce. Move US-16 to Later (D15), re-check at each retro	Elie, with Muhammad

12. Technical decisions

Still open.

Decision	Settled by	When
Build or buy the avatar pipeline	SP-01	End of sprint 1, Fri 2026-11-20 (D4)
Try-on rendering approach	SP-02	End of sprint 1 (D16)
Hosting provider	Muhammad, with the team	Sprint 1
Delivery pipeline tool	Muhammad	Sprint 1
Kiabi access method (API or file feed), garment images, loyalty-card check	OPS-01 answer from Kiabi	Sprint 1 to 2
Retention periods and legal basis	Kiabi privacy contact, with Elie and Muhammad	Sprint 1 (PRIV-01), final in sprint 5 (PRIV-02)
Mirror and tablet hardware	OPS-02	Sprint 4

Proposed for the decision log (to add at the 3a review, after D14). Statuses are Proposed until the team confirms them.

ID	Decision	Evidence	Options set aside, and why	Effect on the backlog
D15	Add the technical items of section 10 and move US-16 to Later	Components have no story, and the 91.5 person-days cover user stories only	Leave the work unplanned: it would surface in sprint 3, which is already full	Eight items added. Roadmap loads change
D16	Version 1 try-on is a 2D overlay from catalog images, adult tops first. Confirmed by SP-02	R8. Glossary fit zones are tops zones	3D garment models: assets and skills we do not have (R6)	US-08 scope and wording

ID	Decision	Evidence	Options set aside, and why	Effect on the backlog
D17	In-store identity: the mirror shows a pairing QR code and the signed-in phone scans it	R10. The NFC tag only counts entries today (brief p.1)	A camera on the mirror: extra hardware. NFC read: needs Kiabi identity access	US-13 wording
D18	The loyalty card is linked once to the account by card number, checked through the Kiabi adapter	Lever L1. The card is rated 7.3 in store and 4.8 online (brief p.4)	NFC read: depends on Kiabi. No link: the same avatar through the card is lost	US-23 added
D19	Photos deleted at creation. The avatar is kept until the shopper deletes it, or 12 months without use (to confirm). Backups kept for a fixed period	R1, R9	Keep data without limit: weak on privacy	US-02, PRIV-01
D20	Size requests are pushed in real time with a polling fallback. The staff tablet uses a shop-level sign-in in the pilot	US-14. An associate must act within seconds	Polling only: slow. Personal staff accounts: needs Kiabi identity integration	US-14 acceptance criteria

13. What changed from version 1

- Added the try-on renderer and the store session service to the application architecture, with matching techniques (sections 2 and 3).
- Replaced the unspecified QR identity step with a pairing flow, and added the loyalty-card link and real-time requests (sections 3 and 6).
- Added the deletion flow, retention table and quality targets (sections 6, 8 and 9).
- Added the Kiabi privacy contact, the pilot shop manager and the shop network (sections 4 and 5).
- Listed the technical backlog items and their capacity effect (section 10).
- Linked the technical risks and added R15 to R18 (section 11).
- Changed status words to the decision log's (Confirmed, Proposed, Open). Version 1 marked D5, D6, D7 and hosting as "Decided" while the log had them as Proposed. The rule-based fit engine is now cited as D7 and D8 (it was cited only as D7).
- Moved the open decisions to section 12 and added D15 to D20.